

QUANTUM COMPUTING

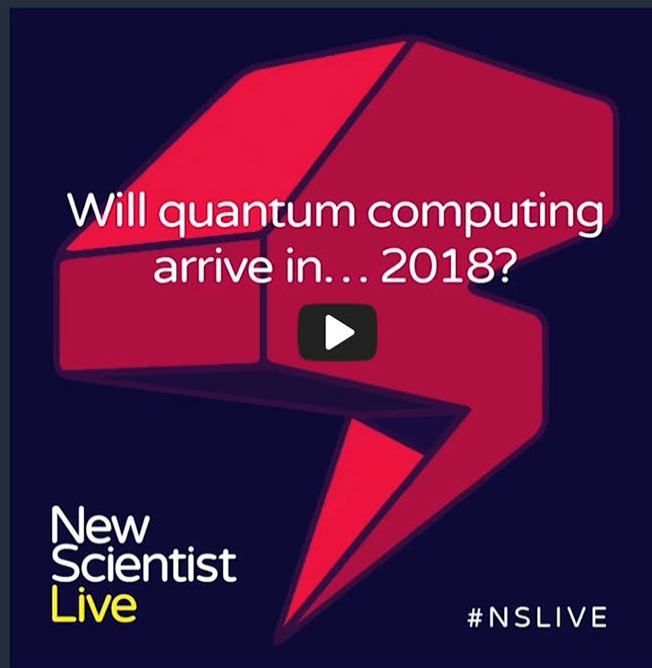
ITS TIME IS NOW (WK1-PT4)

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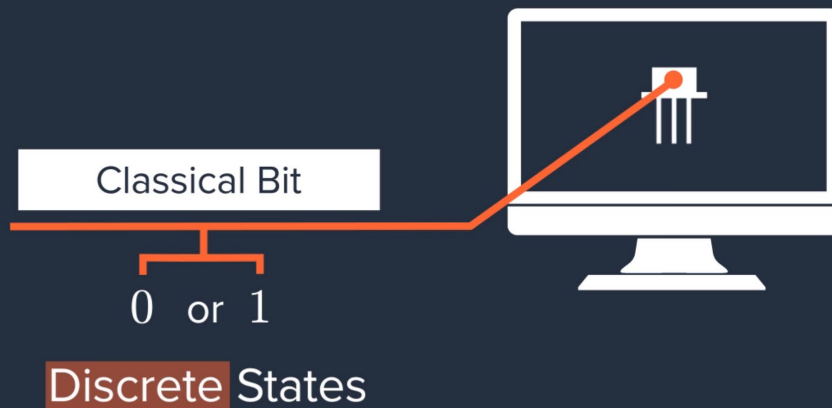
MIT INTRO



@newscilive, 2017

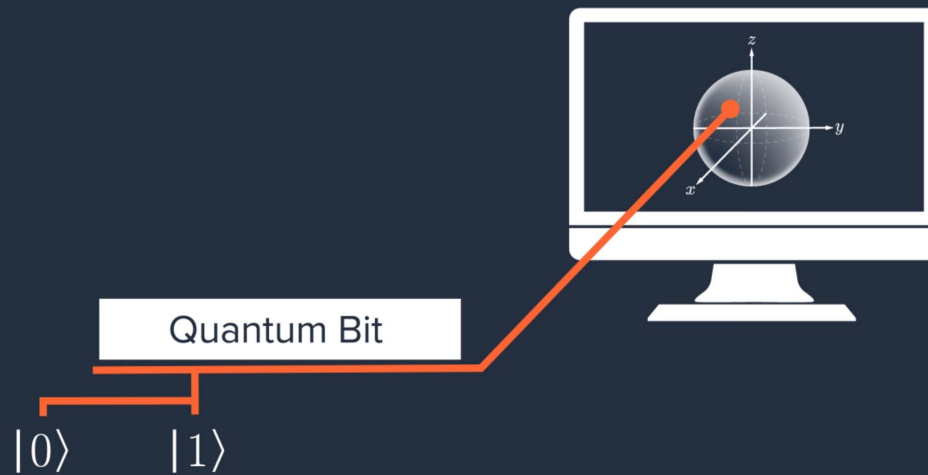
Classical BIT

BIT = Binary Digit



QUBIT

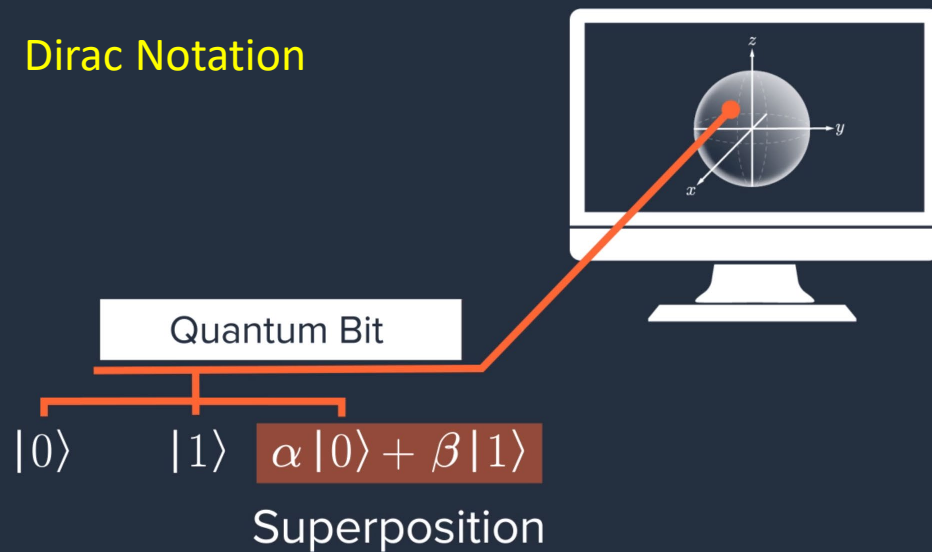
QUBIT = Quantum BIT



QUBIT

Not only represent 0 or 1
But also all positions in between

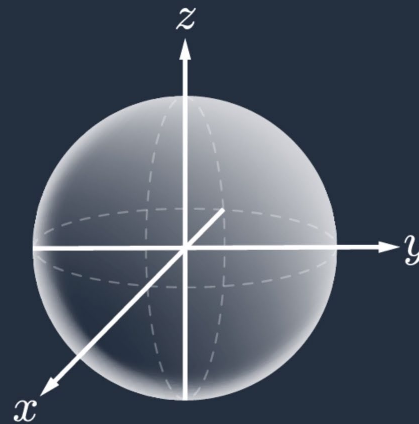
Dirac Notation



BLOCH SPHERE

Bloch Sphere

Three dimensional sphere



EARTH

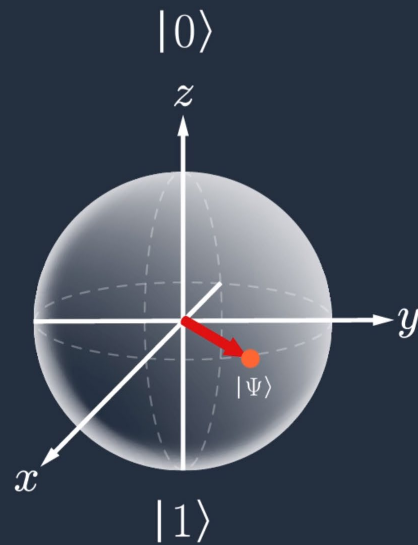
Think of it like the Earth

with 0 at the North Pole
and 1 at the South Pole

There are infinite number
of points on the surface of
the earth



SUPERPOSITION



Superposition

$$|\Psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

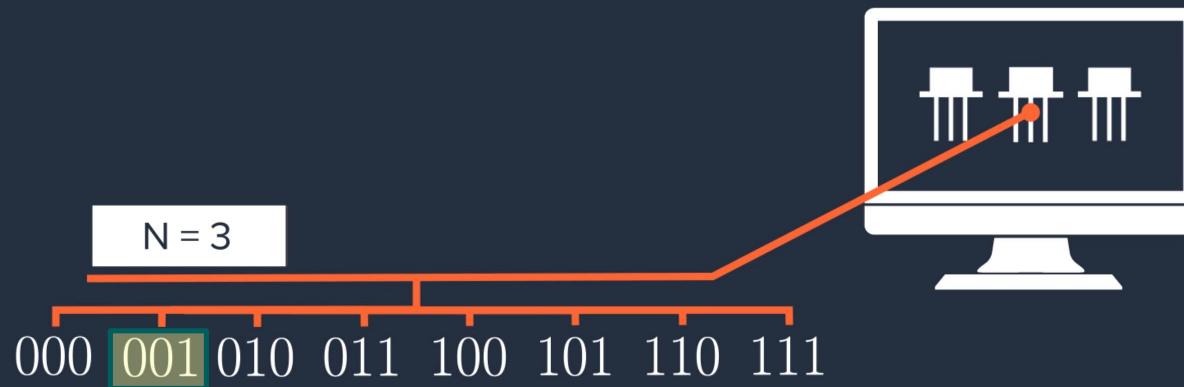
N BITS: ONE N-BIT STATE



N Classical Bits : One N-Bit State

N BITS: ONE N-BIT STATE

If we have 3 Classical BITS or $N=3$



N Classical Bits : One N-Bit State

N BITS: ONE N-BIT STATE

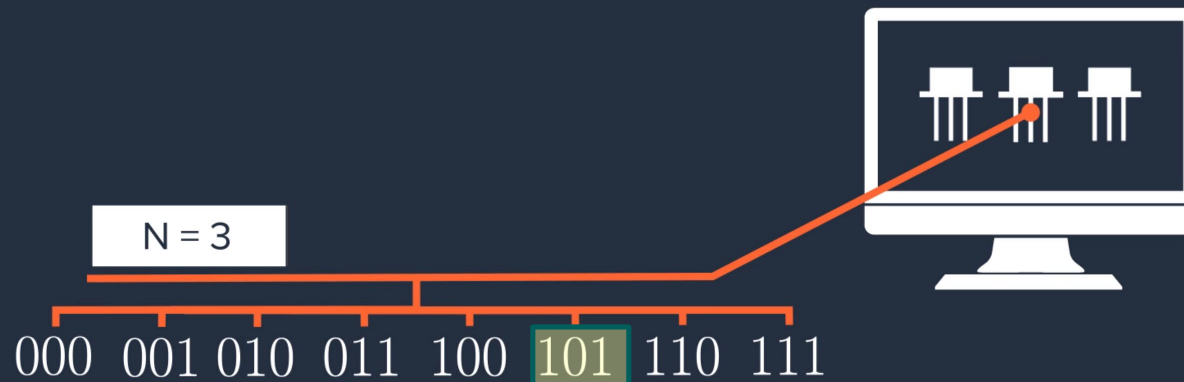
Can only represent one state at a time



N Classical Bits : One N-Bit State

N BITS: ONE N-BIT STATE

Or 101 in this case



N Classical Bits : One N-Bit State

CLASSICAL PROCESS

We process ONE of the states
as an INPUT

And get we get a result
as an OUTPUT

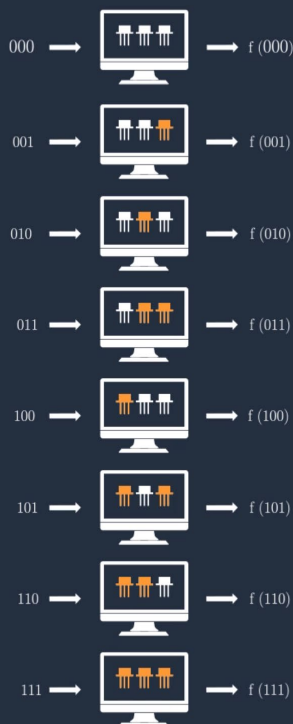


CLASSICAL COMPUTERS

BIG PROBLEMS NEED MORE HARDWARE

If we want to try
DIFFERENT INPUT
States we only have
TWO choices:
MORE HARDWARE

Hardware



CLASSICAL COMPUTERS TRY EVERY COMBINATION

OR TAKE MORE TIME.....



Time

Alan Turing – Break German Enigma



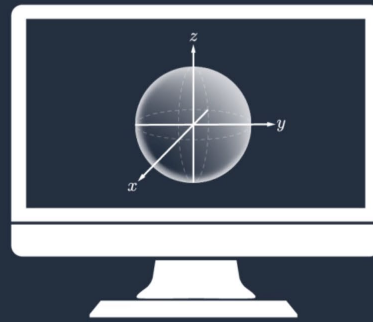
Joseph Desch who was the created
"The Dayton Code Breaking Machine"

United States Naval
Computing Machine Laboratory
At Building 26 of
NCR in Dayton



QUANTUM COMPUTING

Using Superposition
Can represent
simultaneously
ALL this different states



N Qubits : 2^N components in one state

QUANTUM STATE

$$\left. \begin{aligned} &c_1 |000\rangle \\ &+ c_2 |001\rangle \\ &+ c_3 |010\rangle \\ &+ c_4 |011\rangle \\ &+ c_5 |100\rangle \\ &+ c_6 |101\rangle \\ &+ c_7 |110\rangle \\ &+ c_8 |111\rangle \end{aligned} \right\} |\psi\rangle$$



QUANTUM

PARALLELISM & INTERFERENCE

$$\begin{aligned} &c_1 |000\rangle \\ &+ c_2 |001\rangle \\ &+ c_3 |010\rangle \\ &+ c_4 |011\rangle \\ &+ c_5 |100\rangle \\ &+ c_6 |101\rangle \\ &+ c_7 |110\rangle \\ &+ c_8 |111\rangle \end{aligned}$$

Parallelism

$|\psi\rangle \rightarrow$



$\rightarrow |\psi'\rangle$

$$\begin{aligned} &c'_1 |000\rangle \\ &+ c'_2 |001\rangle \\ &+ c'_3 |010\rangle \\ &+ c'_4 |011\rangle \\ &+ c'_5 |100\rangle \\ &+ c'_6 |101\rangle \\ &+ c'_7 |110\rangle \\ &+ c'_8 |111\rangle \end{aligned}$$

Interference